

ARISTO EDUCATIONAL PRESS
HONG KONG DIPLOMA OF SECONDARY
EDUCATION EXAMINATION

(Mock Test 43)

Suggested Solution

Paper 1

- | | | |
|-------|-------|-------|
| 1. B | 11. D | 21. A |
| 2. D | 12. C | 22. C |
| 3. C | 13. B | 23. A |
| 4. D | 14. C | 24. D |
| 5. B | 15. B | 25. A |
| 6. A | 16. A | 26. B |
| 7. D | 17. B | 27. C |
| 8. B | 18. C | 28. C |
| 9. C | 19. D | 29. A |
| 10. A | 20. B | 30. D |

Explanation

1. Answer: B

Macy received a theatre ticket worth \$280 from her father. The \$280 was not paid by Macy herself, so it does not constitute the opportunity cost of choosing to watch the play.

If Macy chooses to watch the play, she has to forgo the \$300 income she could earn from taking a part-time job at a fast-food restaurant. This \$300 is the opportunity cost of choosing to watch the play. At the same time, she also has to forgo the \$200 income she could receive by selling the ticket to her friend. This \$200 is also the opportunity cost of choosing to watch the play.

Therefore, Macy's opportunity cost of choosing to watch the play = \$300 + \$200 = \$500.

2. Answer: D

The word 'should' in Yanny's statement involves a value judgement, so it is a normative statement.

Option A is incorrect. Cody's statement is a positive statement, but the consistency of the statement with real world data is not the reason why it is classified as a positive statement.

Option C is incorrect. Yanny's statement cannot be rejected by facts.

3. Answer: C

(1) is correct. The rules concern who can obtain the tickets, so they are related to the problem of 'for whom to produce'.

(2) is correct. Real-name registration and the non-transferable nature of the tickets directly restrict the holder's right to resell or transfer the tickets to others. Therefore, the buyer's right to transfer over the tickets is weakened.

(3) is incorrect. Buyers still need to pay money (at least \$280) to purchase the tickets, so price competition still exists.

4. Answer: D

Firm A may be a sole proprietorship or a partnership, while firm B is a public limited company.

Option D is an incorrect statement. The profits tax rate for a limited company is higher than that for a sole proprietorship or partnership.

Option A is a correct statement. There is no upper limit on the number of owners for both a partnership and a public limited company. If firm A is a partnership, both firm A and firm B may have 1,000 owners.

Option B is a correct statement. Firm A is not required to disclose its accounting information to the public, whereas firm B must disclose its accounting information to the public.

Option C is a correct statement. Both a sole proprietorship and a partnership lack continuity; if one of the owners goes bankrupt, withdraws, or dies, the firm must be dissolved. A public limited company is a legal entity, so even if one of the owners goes bankrupt, withdraws, or dies, firm B continues to exist.

5. Answer: B

The company has no obligation to redeem shares from shareholders.

Option A is incorrect. The amount of capital raised by issuing shares is not necessarily greater than that raised by issuing bonds.

Option C is incorrect. Since new shareholders may participate in company decision-making, issuing shares will weaken the existing shareholders' control over the company, whereas issuing bonds would not.

Option D is incorrect. Regardless of whether the company issues shares or bonds, existing shareholders can earn capital gains when the share price increases. This is irrelevant to the decision to raise capital through issuing shares and is not an advantage of issuing shares over bonds.

6. Answer: A

(1) and (3) are correct. An ancient poem and free television broadcasts are non-rival and non-excludable in consumption. Their consumption is concurrent in nature, so they are public goods.

(2) is incorrect. Public bus services are excludable in consumption. The bus company can exclude people who have not paid from using the service.

(4) is incorrect. The medicine provided by public hospitals is rival in consumption. When someone obtains some medicine, the quantity available for others decreases.

7. Answer: D

8. Answer: B

9. Answer: C

(1) is correct. Land is a natural resource, and its quantity cannot be increased by human effort.

(3) is correct. Natural wind is a natural resource, while a generator is a man-made resource used in the process of generating electricity.

(2) is incorrect. The return on land is rent, while the return on capital is interest.

10. Answer: A

Average labour productivity of firm A = (30×512) units of output / (30×160) man-hours = 3.2 units of output/man-hour

Average labour productivity of firm B = (25×750) units of output / (25×200) man-hours = 3.75 units of output/man-hour

Firm B's workers have higher productivity, but the percentage of output with quality below standard is also higher. Option A can explain the above difference. Firm B's workers may rush for more output in order to earn more income. In contrast, firm A's workers have no incentive to rush for more output in order to earn more income.

11. Answer: D

(2) and (4) are correct: they do not change as the number of bus services increases.

(1) and (3) are incorrect: they will both increase as the number of bus services increases.

12. Answer: C

Labour (units)	Total output (units)	Marginal product (units)
2	8	-
3	13	5
4	19	6
5	26	7
6	32	6

Option C is correct. When the 6th unit of labour is employed, the marginal product of labour falls from 7 units to 6 units.

Options A and B are incorrect. Since the firm has fixed factors of production (fixed cost is \$10), the firm is operating in the short run. The concepts of economies of scale and optimal scale only apply in the long run.

13. Answer: B

14. Answer: C

Output (units)	Average variable cost (\$)	Total cost (\$)	Marginal cost (\$)
3	4	17	-
4	4	21	4
5	5	30	9
6	6	41	11
7	7	54	13

When output is 6 units (the profit-maximising output), marginal revenue equals marginal cost, which reflects that the market price is \$11. Profit = $6 \times \$11 - \$41 = \$25$

15. Answer: B

A decrease of \$10 in the average cost for each unit of output means that the marginal cost for each unit of output has decreased by \$10. A decrease in marginal cost will lead to an increase in the profit-maximising output level.

Option A is incorrect. A change in fixed cost will not affect the determination of the profit-maximising output level.

Option C is incorrect. When demand decreases, the market price will fall and marginal revenue will fall. With marginal cost for each unit of output unchanged, the profit-maximising output level will decrease.

Option D is incorrect. When marginal cost increases, with marginal revenue for each unit of output unchanged, the profit-maximising output level will decrease.

16. Answer: A

(3) is incorrect. The automobile manufacturer's expansion of its business into the previous production stage cannot expand the market outlet for its products.

17. Answer: B

This situation implies that the prevailing wage rate is below the equilibrium wage rate, resulting in excess demand (shortage) in the labour market. Therefore, the quantity transacted will equal the quantity supplied and will be smaller than the quantity demanded.

18. Answer: C

(2) is correct. For members of fitness centre B, the more times they visit the centre, the lower the average price per visit. As a result, they tend to visit the centre more frequently.

(3) is correct. Since consumers will consume a good up to the quantity at which the price equals the marginal benefit, the price reflects the marginal benefit of the last unit of the good consumed. Members of fitness centre B tend to visit more frequently, and their average price per visit tends to be lower than that of fitness centre A, so the marginal benefit of the last unit of fitness centre service tends to be lower.

(1) is incorrect. Compared with the pricing arrangement of fitness centre B, under fitness centre A's pricing arrangement, after adding the fixed entrance fee, the relative price of a long stay (in terms of a short stay) decreases. According to the law of demand, a larger proportion of users will tend to stay for a long time rather than for a short time. On average, users will stay for a longer time.

19. Answer: D

The canned coffee of other brands is a substitute for brand A's canned coffee. When more firms enter the canned coffee market, the demand for brand A's canned coffee will decrease. When the price of coffee beans rises, the production cost of brand A's canned coffee will increase and its supply will decrease.

20. Answer: B

When the price of replacement parts for smartphones decreases, the cost of providing smartphone repair services will decrease and its supply will increase. The equilibrium point will shift from E_1 to E_2 .

Option A is incorrect. When the price of brand-new smartphones decreases, if a smartphone breaks down, people will be more likely to buy a new smartphone rather than repair the old one. Therefore, the demand for smartphone repair services will decrease.

Option C is incorrect. When the equilibrium wage of repair technicians of other electronic products increases, the supply of smartphone repair technicians will decrease and the supply of smartphone repair services will decrease.

Option D is incorrect. When smartphones break down more frequently, the demand for smartphone repair services will increase.

21. Answer: A

When the supply of good X decreases, the equilibrium price of good X will increase. If good X and good Y are substitutes, the increase in the equilibrium price of good X will lead to an increase in the demand for good Y. The quantity demanded of good Y will increase at all given prices.

22. Answer: C

The government setting up more free Wi-Fi hotspots will increase the availability of substitutes, causing the price elasticity of demand for mobile data services to increase.

23. Answer: A

If good X and good Y are in joint supply, when the demand for good X increases, the equilibrium price of good X will increase and the supply of good Y will increase. The price of good Y will decrease and the quantity demanded will increase.

Percentage change in quantity demanded = $(300 - 250) / [(250 + 300) / 2] \times 100\% = 18.18\%$

Since the percentage change in price (25%) is higher than the percentage change in quantity demanded (18.18%), the demand for good Y is inelastic.

24. Answer: D

Total revenue net of tax = $(\$5 - \$2) \times 10 = \$30$

Option A is incorrect. If the government imposes a price ceiling of \$3 on good X, the quantity transacted of good X will equal the quantity supplied (10 units).

Option B is incorrect. If the government imposes a price floor of \$7 on good X, the quantity transacted of good X will decrease from 10 units to 8 units.

Option C is incorrect. After providing a per-unit subsidy, the equilibrium price of good X will remain at \$5.

25. Answer: A

(1) is correct. If the price of drone motors falls, the production cost of drones will decrease and the supply will increase. The quota of Q_1 remains effective, so the quantity transacted will remain unchanged.

(2) is correct. If drones become more popular, the demand for drones will increase. The price will rise while the quantity transacted remains unchanged. Total sales value will increase.

(3) is incorrect. If the government increases the quota, the market price will decrease.

26. Answer: B

27. Answer: C

28. Answer: C

Option B is incorrect. When the price is above the equilibrium price, a surplus occurs. The quantity transacted (= quantity demanded) is smaller than the equilibrium quantity (= the efficient quantity, assuming no externalities). Underproduction will occur.

Option D is incorrect. When the quantity demanded is greater than the quantity supplied, a shortage occurs. The quantity transacted (= quantity supplied) is smaller than the equilibrium quantity (= the efficient quantity, assuming no externalities). Underproduction will occur.

29. Answer: A

The price ceiling of P_1 is below the equilibrium price and is an effective price ceiling.

(1) is correct. Before setting the price ceiling at P_1 , consumer surplus is zero. After setting the price ceiling at P_1 , consumer surplus is area P_1bcd .

(2) is correct. Before setting the price ceiling at P_1 , producer surplus is area abE_0 . After setting the price ceiling at P_1 , producer surplus is area aP_1d . Producer surplus decreases by area P_1bE_0d .

(3) is incorrect. The area by which producer surplus decreases is greater than the area by which consumer surplus increases. Total social surplus will decrease, and the deadweight loss is area dcE_0 .

30. Answer: D

The Gini coefficient of Country A based on post-tax post-social transfer household income is lower than that based on original household income. This means that household income after tax and social transfer is more evenly distributed than original household income.

Option A is incorrect. The data does not reflect the actual amounts of tax revenue and social transfer.

Option B is incorrect. The Gini coefficient based on original household income in 2025 is higher than that in 2020, which means that the Lorenz curve plotted using original household income in 2025 is farther away from the line of perfect income equality than in 2020.

Option C is incorrect. The data does not reflect changes in the number of households in different income groups.

Paper 2

Section A (30 marks)

1. No. (1 mark)
- Obtaining the scholarship for admission to the master's programme would increase the value of the third option. If the value of the third option increases above that of the second option, to study a master's degree programme would become the highest-valued option forgone. The opportunity cost for choosing to work as an administrative officer in the government would change from the value of working as a junior accountant in an accounting firm to the value of studying a master's degree programme. (2 marks)
2. (a) Listed company / Public limited company. (1 mark)
- A change:
- The shares of Hang Seng Bank Limited will no longer be freely traded on the stock exchange.
 - Any reasonable answer
- (Mark the **FIRST** point only)
- (b) After the announcement, people expected that HSBC Holdings plc would acquire the shares of Hang Seng Bank at \$155. This expected price increase would have led to an increase in the demand for Hang Seng Bank's shares. (2 marks)
3. (a) An example of a variable cost:
- Wages of part-time employees
 - Electricity/water charges
 - Maintenance expenses for facilities
 - Any reasonable answer
- (Mark the **FIRST** point only)
- This is because these costs will decrease when part of the recreational facilities is closed. (1 mark)
- (b) Capital, because they are man-made resources used in the production process. (2 marks)
- (c) Division of labour / Specialisation. (1 mark)
- Reasons:
- Choosing the best people to do the job
 - Practice makes perfect
 - Saving time on training
 - Saving time on moving from one task to another
 - Stimulus to mechanisation
 - Any reasonable answers
- (Mark the **FIRST THREE** points only)
4. (a) If demand is less elastic than supply, consumers' tax burden will be greater than producers' tax burden. (1 mark)

(b) Verbal explanation:

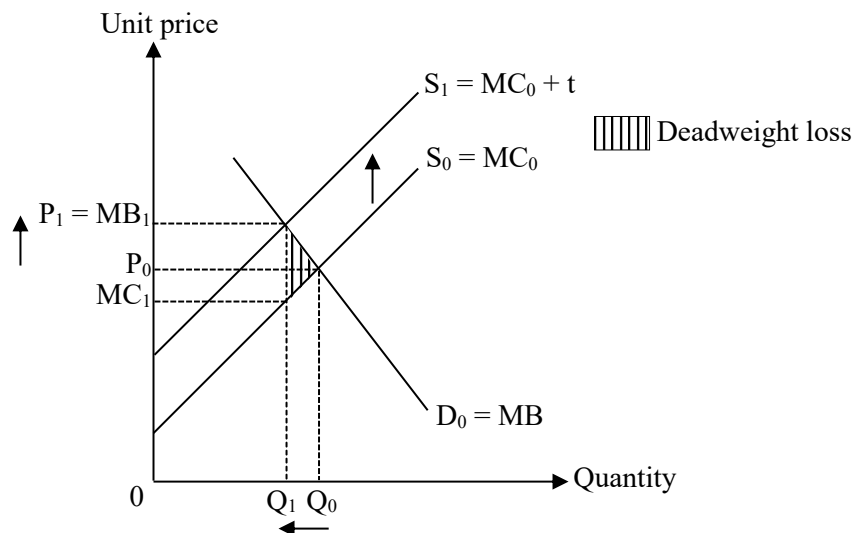
- After the government imposes a per-unit sales tax on good X, the supply of good X will decrease. (1 mark)

- The marginal benefit of the last unit will be higher than the marginal cost / there will be a deadweight loss, and economic efficiency will worsen. (1 mark)

Illustrate in the diagram:

- upward shift of the supply curve (1 mark)

- correct position of the deadweight loss / marginal benefit higher than marginal cost (1 mark)



5. (a) Providing these courses involves the use of scarce resources which have alternative uses. / Some people would prefer to have more of these courses. (1 mark)

(b) These courses can help equalise opportunities, as unemployed people can receive more education, thereby improving their productivity and income earning power in the future. (3 marks)

6. Verbal explanation:

- A sharp decrease in the cocoa bean harvest would cause the price of cocoa beans to increase. As a result, the production cost of chocolate would increase and its supply would decrease. (1 mark)

- In Country A, chocolate is a normal good. During an economic recession, people's average income decreases, leading to a decrease in the demand for chocolate. (1 mark)

- When the decrease in demand is greater than the decrease in supply, the equilibrium price will decrease. (1 mark)

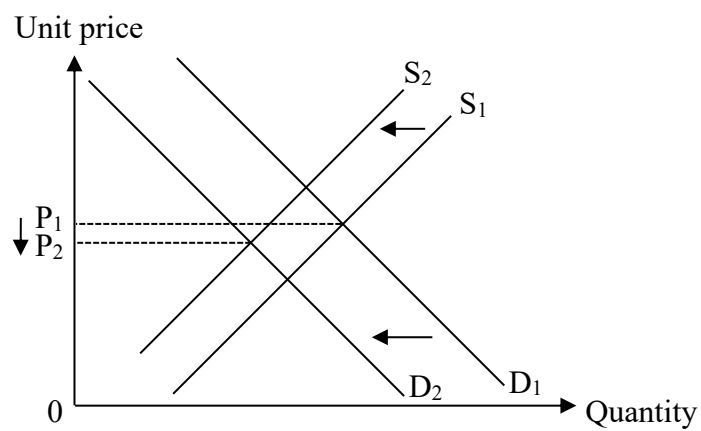
Illustrate in the diagram:

- leftward shift of the supply curve
- leftward shift of the demand curve
- a decrease in the equilibrium price

(1 mark)

(1 mark)

(1 mark)



Section B (40 marks)

7. (a) Tertiary production, because they provide property management services. (2 marks)
- (b) Possible reasons:
- Supervisory and technical workers may possess more human capital than clerical and secretarial workers.
 - Supervisory and technical workers work longer hours than clerical and secretarial workers.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- (c) The occupational mobility of supervisory and technical workers is lower than that of clerical and secretarial workers, because their skills are more specialised than those of clerical and secretarial workers, making it relatively more difficult for them to transfer to other industries. (3 marks)
8. (a) Oligopoly. (1 mark)
- Features:
- There are only a few dominant sellers in the market.
 - Firms are interdependent in pricing strategies.
 - There are various barriers to entry for new firms.
 - Neither buyers nor sellers have perfect information about the market.
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)
- (b) The loan services provided by banks and the loan services provided by finance companies are substitutes. Issuing more finance company licenses will increase the number of substitutes. As a result, the price elasticity of demand for the loan services provided by banks will increase. (3 marks)
- (c) Vertical backward integration, (1 mark)
- because the credit reporting agency provides credit reports to banks, which are inputs required for banks to provide loan services. (1 mark)
- Possible motives:
- to have better control over the quality of the credit reports (inputs)
 - to ensure a reliable supply of credit reports (inputs)
 - Any reasonable answers
- (Mark the **FIRST TWO** points only)

2@, max: 4 marks

1@, max: 2 marks

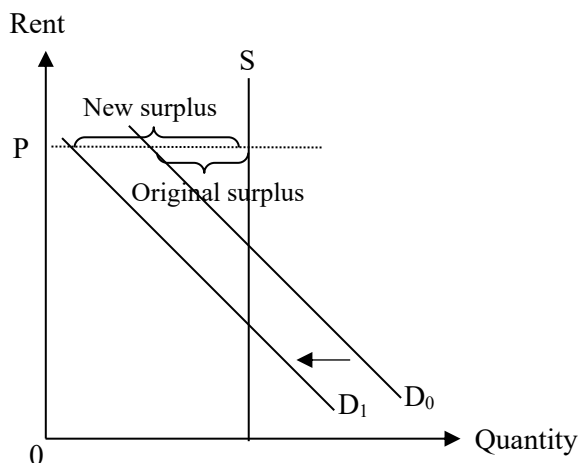
1@, max: 2 marks

9. (a) Verbal explanation:

Elderly people have a lower incentive to consume. Population ageing causes merchants' expected profit to decrease, leading to a decrease in the demand for shop premises. At the given rent level, the excess supply of shop premises increases, i.e. the vacancy rate rises. (2 marks)

Illustrate in the diagram:

- a vertical supply curve and a downward-sloping demand curve (1 mark)
- rent is set above the equilibrium level (1 mark)
- leftward shift of the demand curve (1 mark)
- surplus (excess supply) increases (1 mark)



- (b) Buying a large quantity of goods and buying a small quantity of goods both incur the same transportation cost / time cost. When residents travel to more distant areas for shopping, after adding the same transportation cost / time cost, the relative price of purchasing a large quantity of goods (in terms of purchasing a small quantity) will decrease. According to the law of demand, residents will tend to buy more goods. As a result, the average number of items purchased by residents on each shopping trip will increase. (3 marks)
- (c) The value of the highest-valued option forgone (i.e., renting out the shop premises) will decrease. Therefore, the opportunity cost for the owner of using the shop premises to operate her own business will decrease. (2 marks)
- (d) Marks awarded for effective communication (EC: max 1 mark)

Marks	Performance
1	• Arguments presented with some support of the source/data and economic theories • Some irrelevant materials presented • Answers presented in a less organised way with some repetition of ideas • A clear message conveyed with some inappropriate use of language/words/terms/symbols
0	• Arguments presented with no support of the source/data and economic theories • Materials unrelated to the gist of the question presented • Inconsistent arguments presented • Limited ideas expressed with inappropriate use of words/terms/symbols

The maximum marks for content are 9 marks. Answers may include the following:

Economic efficiency

- The subsidy would increase consumers' demand for goods. The price and quantity transacted of goods will increase. The producer surplus will increase. As consumers receive a subsidy from the government, the price they actually pay decreases, while the quantity transacted increases, resulting in an increase in the consumer surplus. However, the increases in the consumer surplus and producer surplus will be smaller than the total amount of the subsidy, resulting in a deadweight loss. Economic efficiency will worsen.

The Gini coefficient based on post-tax post-social transfer household income

- Since the subsidy is calculated as a fixed percentage of the consumption amount, if low-income and high-income groups have the same consumption amount, the subsidy received by the low-income group as a percentage of their income will be higher. The Gini coefficient based on post-tax post-social transfer household income will decrease.
- Since the subsidy is calculated as a fixed percentage of the consumption amount, if the high-income group has a higher consumption amount than the low-income group, the subsidy received by the low-income group as a percentage of income may be higher than, lower than, or equal to that of the high-income group. Therefore, the change in the Gini coefficient based on post-tax post-social transfer household income is uncertain.

The vacancy rate of shop premises in shopping malls

- After the subsidy is provided, consumers' demand for goods will increase. More firms will rent shop premises to run a business, causing the vacancy rate of shop premises in shopping malls to decrease.
- If people expect the subsidy to be temporary or one-off, and renting shop premises to operate a business involves fixed costs, they may not enter the market. As a result, the extent of the decrease in the vacancy rate will be limited.